

SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I K Sumanth Kumar Reddy (192425222) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Sumanth

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
The system must handle millions of concurrent users with minimal buffering.
Design a cloud architecture using scalability and caching techniques.
Explain how load balancing and auto-scaling can ensure smooth streaming.
Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
They face challenges in maintaining consistency and managing deployments.
Propose a solution using container orchestration and multi-cloud strategies.
Explain how service discovery and scaling can be handled efficiently.
Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
The organization needs a robust backup and disaster recovery strategy.
Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements.
Design a hybrid architecture that ensures secure communication between environments.
Explain how identity management and encryption should be implemented.
Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down.
Propose a multi-region deployment strategy in the cloud.
Explain how data replication and failover mechanisms should be configured.
Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

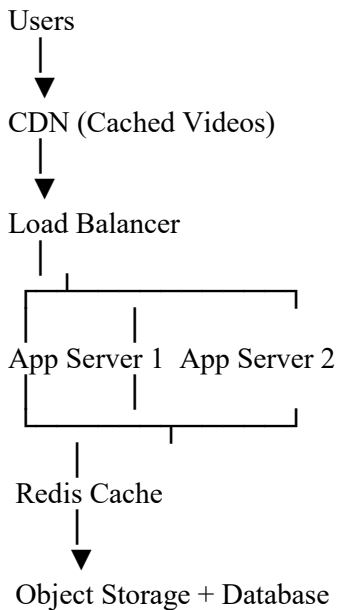
1. Media Company – Cloud Video Streaming Platform

a) Cloud Architecture Design

Architecture:

- Users access the platform through a **Content Delivery Network (CDN)**.
- CDN caches videos at edge locations.
- Requests are routed to a **Load Balancer**.
- Load balancer distributes traffic among multiple application servers.
- Application servers run in **Auto Scaling Groups**.
- Video files are stored in **Cloud Object Storage**.
- Metadata is stored in a managed database.
- Frequently accessed data is stored in an in-memory cache (Redis/Memcached).

Architecture Flow



b) Load Balancing and Auto Scaling

Load Balancing

- Distributes incoming traffic equally among servers.
- Prevents server overload.
- Improves response time.
- Removes failed servers automatically.

Auto Scaling

- Automatically adds servers during peak traffic.
- Removes extra servers during low traffic.
- Ensures uninterrupted video streaming.
- Minimizes buffering.

c) Cost Optimization Strategies

- Use **CDN** to reduce bandwidth costs.
- Store old videos in cheaper archive storage.
- Enable auto scaling to avoid paying for idle servers.
- Use reserved instances for predictable workloads.
- Compress videos using efficient codecs.
- Monitor cloud costs regularly.

2. Fintech Startup – Multi-Cloud Container Deployment

a) Proposed Solution

Use:

- Docker containers
- Kubernetes for orchestration
- Multi-cloud deployment (AWS + Azure + GCP)
- CI/CD pipeline
- Container Registry

Architecture:

Developer



CI/CD Pipeline



Docker Image



Kubernetes Cluster



b) Service Discovery and Scaling

Service Discovery

- Kubernetes DNS automatically finds services.
- Services communicate without knowing IP addresses.

Example:

Payment Service



Authentication Service

Scaling

- Horizontal Pod Autoscaler (HPA)
- Cluster Autoscaler
- Load balancing between pods

Scaling depends on:

- CPU usage
- Memory usage
- Request count

c) Benefits and Risks

Benefits

- High availability
- Avoid vendor lock-in
- Better disaster recovery
- Flexibility
- Improved performance

Risks

- Higher management complexity
- Increased security challenges
- More networking costs
- Difficult monitoring
- Data synchronization issues

3. E-Learning Platform – Backup and Disaster Recovery

a) Backup Solution

Use:

- Cloud-native backup service
- Object storage versioning
- Cross-region replication
- Database snapshots
- Scheduled backups

Architecture

Primary Storage

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Versioning

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Daily Backup

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Cross Region Replication

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Disaster Recovery Site

b) Achieving RTO and RPO

RTO (Recovery Time Objective)

Time required to restore the application.

Achieved by:

- Automated failover
- Pre-configured backup servers
- Fast recovery procedures

Example:

Target RTO = 15 minutes

RPO (Recovery Point Objective)

Maximum acceptable data loss.

Achieved by:

- Continuous replication
- Frequent backups
- Database synchronization

Example:

Target RPO = 5 minutes

c) Business Continuity Steps

1. Perform automatic daily backups.
2. Enable versioning.
3. Replicate data to another region.
4. Test disaster recovery regularly.
5. Monitor backup success.
6. Use automated failover.
7. Maintain disaster recovery documentation.

4. Government Agency – Hybrid Cloud

a) Hybrid Cloud Architecture

Sensitive data:

- Stored on-premises

Non-sensitive applications:

- Hosted in public cloud

Secure communication:

- VPN
- Dedicated private connection

Architecture

On-Premises Data Center



VPN / Dedicated Link



Public Cloud



Application Servers

b) Identity Management and Encryption

Identity Management

- Single Sign-On (SSO)
- Multi-Factor Authentication (MFA)
- Role-Based Access Control (RBAC)
- Identity Federation

Encryption

Data at Rest:

- AES-256 encryption

Data in Transit:

- TLS/SSL encryption

Use Key Management Service (KMS) for encryption keys.

c) Challenges

- Complex infrastructure management
- Data synchronization
- Security policy consistency
- Network latency
- Compliance requirements
- Higher operational costs

5. SaaS Provider – High Availability and Fault Tolerance

a) Multi-Region Deployment Strategy

Deploy the application in multiple cloud regions.

Architecture

Users



Global Load Balancer



Region A



Region B



App



App



Database Replication



If Region A fails, traffic is automatically routed to Region B.

b) Data Replication and Failover

Data Replication

- Synchronous replication for critical databases.
- Asynchronous replication for backup regions.
- Continuous database synchronization.

Failover

- Automatic health checks.
- DNS failover.
- Load balancer redirects traffic to healthy regions.
- Backup databases become active automatically.

c) Monitoring and Alerting

Use cloud monitoring tools to monitor:

- CPU utilization
- Memory usage
- Disk usage
- Network traffic
- Application response time
- Database performance
- Error rates

Configure alerts through:

- Email
- SMS
- Mobile notifications
- Incident management systems